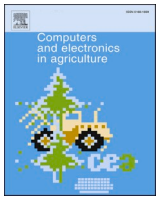
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Towards real-time weed detection and segmentation with lightweight CNN models on edge devices

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ABSTRACT

With the resurgence of on-the-go site-specific weed control technologies, research has increasingly focused on real-time weed identification using computer vision and machine learning. While Convolutional Neural Networks (CNNs) have been successfully used for automatic weed detection, many studies prioritize accuracy over inference speed and computational resource demand. The objective of this work was to develop, deploy, and evaluate CNN-based computer vision models on computationally resource-restrained edge devices for real-time weed classification and segmentation. RGB images were collected by UAVs over corn and soybean fields infested with Palmer Amaranth (*Amaranthus palmeri*). A few object detection (OD) and semantic segmentation (SS) models were modified, if necessary, to be lightweight, then trained and deployed on edge devices for real-time weed detection with resized input images in dimensions of 256 × 256 pixels. Specifically, the original backbones of Single-Shot Detector (SSD) and DeepLabv3+ models were replaced by a pre-trained MobileNetV3 backbone and deployed on a Jetson Nano. A YOLOv8n model and a proposed MobileNetV4-Seg model were trained and deployed on a Jetson Orin Nano. All models achieved real-time performance with acceptable accuracies with the data collected in corn and soybean fields. Among them, the MobilenetV4-Seg achieved the best performance, with IoUs of 69.9 % and 76.8 %, F_1 scores of 82.3 % and 86.9 %, for the corn and soybean datasets, respectively, and 44 FPS on the Jetson Orin Nano. Overall, this study demonstrated a use case and feasible workflow for developing and deploying lightweight CNN models on resource-constrained edge devices for on-the-go real-time site-specific applications.

1. Introduction

Weeds are the rivals of crops that fight for the air, water, soil nutrients, and space. To ensure optimum crop yield, it is important to control the weeds. Based on the research conducted by the Weed Science Society of America on the corn and soybean fields across North America spanning from the year 2007 to 2013, the annual average corn yield loss could reach 52 % and the average soybean yield loss could be 49.5 %, resulting in combined potential losses of $43 billion if weed is left untreated (Soltani et al., 2016, 2017). Numerous mechanical, biological, and chemical weed control measures have been used by farmers. Mechanical weed control techniques include cultivation prior to seeding, inter-row cultivation, mowing, burning and flame weeding, mulching, tilling, hoeing, and hand removal. Biological weed control entails introducing natural destroyers of the target weeds, such as insects, mites, and pathogens, aiming to reduce and maintain its population to an acceptable level. The most commonly and widely used approach for weed management is chemical control, such as performing uniform broadcast herbicide spraying as it is more efficient compared to the mechanical and biological weed controls. However, the major drawbacks of uniform rate herbicide application include the increased chemical footprints in the environment and the evolution of herbicide-resistant weeds (Nath et al., 2024). Other limitations are health hazards, potential damage to non-target species, and related economic loss. Because of this, the concept of Site-Specific Weed Management (SSWM) was brought as a weed control method that considers the spatial

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variability and temporal dynamics of weed populations in the field. It precisely targets individual weeds or patches of weeds to help reduce herbicide use, increase biodiversity, and reduce the chances of herbicide crop injury.

With the resurgence of on-the-go, site-specific weed control machinery and technologies in recent years, there has been a surge of interest in research focused on the rapid and precise identification of weeds in real-time, leveraging cutting-edge computer vision and machine learning algorithms. A backbone technology enabling automatic SSWM is computer vision-based weed identification. Traditionally, weed pressure maps are derived first from data collected from remote sensing platforms such as satellites and manned or unmanned aerial and ground vehicles (Wu et al., 2021) before targeted weeding or variable-rate spraying. Before the widespread adoption of Convolutional Neural Networks (CNNs), various image processing and machine learning based computer vision algorithms were employed to develop weed detection models. In these works, different visual characteristics have been used for weed detection under the wide range of agricultural field conditions. Extracting features for implementing the conventional supervised machine learning models for weed detection is naturally challenging due to the similar appearances of weeds and crops. CNN-based Deep Learning (DL) models are a preferred choice compared to the traditional supervised machine learning models and the conventional unsupervised machine learning models. This is due to the fact that, CNNs perform joint feature and classifier learning, where the feature extraction is learned automatically through deep learning models rather than handled separately before applying a classifier; and the classifier is learned simultaneously with feature extraction (Gu et al., 2018).

Object detection (OD) (Zhao et al., 2019) and semantic segmentation (SS) (Liu et al., 2019) are two main categories of CNN-based detection approaches for weed detection that have been largely investigated and adopted in recent years. A typical architecture of an OD model consists of a backbone, Region Proposal Network, and a classification head. It is less computationally complex and hence requires relatively less resources but provides faster inference speeds. A typical architecture of a SS model is built with a backbone and an encode-decoder structure. SS models are often computationally more intensive and, hence, require more resources and come with a lower inference speed compared to the OD models. The OD model outputs a bounding box around the detected object whereas the SS model performs the classification for every pixel presented in the image and outlays polygon segmentation masks along the detected object's periphery. Hence, in the application of weed detection, OD models are often used for species classification and counting (Norouzzadeh et al., 2017), while SS models are often used more for precise patch contour delineation and density estimation (Asad and Bais, 2020). A lot of the previous studies on the OD and SS model based weed detections were deployed on resourceful desktop or laptop computers. In one such work, Goyal et al. (2025) utilized Mask RCNN and YOLOv8 for weed detection in complex, highly occluded potato fields during the post-emergence stage. YOLOv8 achieved a mAP@0.5 of 83.4 %, while Mask RCNN achieved 79.0 %. Mask RCNN showed higher precision and recall for the weed class. The testing was conducted on a computer with a Tesla T4 GPU. Rehman et al. (2024) introduced a novel weed detection model for soybean crops by modifying YOLOv5 architecture, incorporating Ghost Convolution, BottleNeckCSP, and Efficient Channel Attention (ECA) layers to enhance feature extraction and weed identification accuracy from drone imagery. They achieved a precision of 72.5 %, recall of 68.0 %, and mAP@0.5 of 73.9 %, outperforming existing models like RT-DETR and YOLOv10 using NVIDIA TITAN Xp GPU. In another work, Xu et al. (2023) presented a novel approach by combining the visible color index with the instance segmentation method for weed detection in soybean using ResNet101_v and DSASPP encoder and decoder architectures. Experimental results on soybean field imagery showed that the proposed approach achieved the overall performance of a mean IoU (mIoU) of 93.9 % and an average 1.8 FPS with input images in 1472 × 1120 pixels using a workstation computing system with a NVIDIA GeForce GTX 1080 GPU. Habib et al. (2024) developed a new deep learning segmentation model DWUNet, and compared it to other SOTA models (SharpUNet, SegNet, DCANet, MobileNetv2UNet, MobileNetv3UNet) using a dataset of pea crops images. DWUNet achieved a Jaccard index of 0.825 and an inference speed of 8 ms per image on a Tesla T4 GPU. DWUNet demonstrates superior detection accuracy and generalization capability compared to state-of-the-art models. Bhatti et al. (2024) used a CNN-based image segmentation model with a U-Net architecture to segment agricultural imagery for crop and weed identification. The study explored the impact of training data split ratio on model performance. Key findings emphasized the importance of training data size, with larger datasets leading to superior segmentation performance (77.4 % IoU and 83.5 % F_1 score). Although they did not specifically mention the configuration, they stated that they required high-performance GPU for their study.

With the advancement of computational hardware and computer vision techniques in recent years, on-the-go sensing, decision-making, and application have become true. There are a few featured products on the market. For example, John Deere Corporation (Moline, Illinois, USA) introduced their See & Spray series of self-propelled sprayers in 2017 utilizing computer vision and machine learning to precisely apply herbicides only where weeds are present. The onboard computers (i.e., edge devices) on the sprayers perform real-time inferences with up to 12 mph of a vehicle's moving speed. Other similar technologies and sprayer systems that are well known on the market include but are not limited to the One Smart Spray system by Bosch and BASF, the selective spraying technology by Greeneye Technology, the Spot Spraying system by Agrifac, and the ARA targeted spraying by Ecorobotix. They are all self-propelled sprayers or add-on systems.

Other than commercial systems, researchers also successfully implemented OD and SS models on various types of edge devices. When there is less limitation on system payload and power consumption, the selection of the edge devices can be focused on maximizing the real-time inference accuracy. This is often the case with the built-in computers or third-party computing platforms on unmanned ground-based vehicle platforms. Li et al. (2024) reproduced SOTA models (R-CNN, YOLO, Transformer) to validate the new winter wheat weed dataset and then improved YOLOv8 and DINO using spatial attention mechanism (SAM), Non-local block (NLB) and deformable convolution network v2 (DCNv2). Results showed that NLB improved YOLOv8 and DINO performance, with YOLOv8_NLB achieving 88.3 % mAP, and DINO_NLB reaching 89.4 % mAP. For real-world deployment, YOLOv8_NLB was converted to TensorRT format for Jetson Xavier NX, with the TensorRT_FP16 model having 40.5 FPS and 86.7 % mAP. Rai et al. (2024) deployed different YOLO-Spot models on an NVIDIA Jetson AGX Xavier with 384 × 384 pixel input images, and achieved an accuracy of 84.0 % and a 13.8 FPS with the YOLO Spot M model for the identification of four weed species, namely, kochia, ragweed, horseweed, and redroot pigweed in corn and soybean. Rai & Sun (2024) developed a single-stage deep learning, YOLO-inspired architecture WeedVision, for weed detection and instance segmentation in drone-acquired images. They evaluated its performance on both a desktop GPU and an edge device, AGX Xavier. Model conversion to TorchScript and ONNX formats was performed for edge deployment. The model achieved 85.4 % bounding-box precision and 82.8 % masking precision on the GPU. The converted TorchScript version also performed well and provided real-time weed detection on the edge device.

Deployment of OD and SS-based weed detection models on relatively resource-constrained edge devices has also been investigated. In one such work, Farooq et al. (2022) used the YOLOv4-tiny OD model for weed detection and a mAP@0.5 up to 85.55 %. This model was converted to TensorFlowLite and deployed on a Raspberry Pi 4 (model B) and a 4 FPS inference speed was achieved for input images of 416 × 416 pixels. They used TensorRT as a conversion method for the model to be edge-device deployable. Harders et al. (2023) evaluated the

performances of three lightweight deep learning models, i.e., YOLOv4-tiny, YOLOv4-tiny-3l, and YOLOv7-tiny, on three edge devices, i.e., NVIDIA Jetson Nano, TX2, and Xavier NX. For Jetson Nano which has the least computational power among the three, they achieved F_1 scores of 69.97 %, 74.30 %, and 68.55 %; and inference speed of 15.1, 13.3, and 11.8 FPS, for YOLOv4-tiny, YOLOv4-tiny-3l, and YOLOv7-tiny, respectively, with input images of 480 × 480 pixels and an FP32 quantization. For SS, DeepLabV3 SS model was deployed on the NVIDIA Jetson Nano by Assunção et al. (2022)) and achieved 64.0 % mIoU with 5.9 FPS for input images of 1296 × 966 pixels.

Despite the popularity and wide adoption of UAVs (commonly known as drones) among farmers for sensing and spraying in many parts of the world, few commercial UAV systems, if there are any, and little research work can be found on real-time onboard sensing and decision-making for specific agricultural applications such as weed detection. The key limiting factors here are the payload and power supply of the UAV system, while the fast mobile velocity of the UAVs compared to some other cases requires a very fast inference speed for real-time applications. Compact and lightweight edge devices with lower power consumption are needed without sacrificing too much inference speed and accuracy. In this study, we used the NVIDIA Jetson Nano and Jetson Orin Nano edge devices due to their availability when we conducted this study and their compactness and less power consumption. Jetson Nano used to be the lowest computational performance compared to the other NVIDIA Jetson family members. Specifically, the Jetson Nano has 472 giga floating-point operations per second (GFLOPS), Jetson TX2 has 1.33 tera floating-point operations per second (TFLOPS), Jetson Xavier NX has 21 TFLOPS, and Jetson AGX Xavier has 32 TFLOPS. Jetson Nano board and development kit have the lowest weights in the series and the lowest power consumption (10 W) compared to NVIDIA Jetson Orin Nano, TX2, Xavier NX, and AGX Xavier's power consumptions (15 W, 15 W, 20 W, and 30 W, respectively) (NVIDIA Developer, 2024). If an acceptable benchmark model performance can be achieved with the currently available Jetson Nano and Orin Nano systems, superior performance can be achieved with upgraded edge computing devices. In addition, as we discussed earlier, the type of the CNN models, i.e., the OD versus the SS models, also have different applications in weed detection and impact on real-time applications. Hence, the overall objective of this study was to develop and compare CNN-based OD and SS models that can be deployed on the resource-constrained edge devices (e.g., companion computer for the spraying UAV) for real-time weed identification. The specific objectives were:

(1) To modify example off-the-shelf OD and SS models into lightweight versions for real-time weed detection or segmentation on resource-constrained edge devices.
(2) To deploy the lightweight OD and SS models on two resource-constrained edge devices (i.e., Jetson Nano and Jetson Orin Nano) and evaluate their detection and segmentation performance, as well as inference speed.

2. Materials and methods

2.1. Data collection and preparation

In this study, we targeted a major weed in row crop production in North America – the Palmer Amaranth (*Amaranthus palmeri*). RGB images of Palmer Amaranth were collected in a commercial corn field in Carleton, Nebraska, USA, on July 1st, 2021, using a sensing UAV (Phantom 4 RTK, DJI, Shenzhen, China) and a built-in camera. Similarly, RGB images were collected in a soybean field in South Central Agricultural Lab (SCAL), Clay Center, Nebraska, USA, on July 18th, 2022, using another sensing UAV, which was the latest model then (Matrice 300 RTK, DJI, Shenzhen, China), and a compatible plug-and-play high-resolution camera (Zenmuse P1, DJI, Shenzhen, China) (Table 1). Palmer amaranth plants were about 1.2 to 1.5 feet tall at the time of data collection in 2021; and about 1.8 to 2.2 feet tall in 2022. The UAVs were manually flown at 10 m and 5 m above ground level (AGL) in order to bypass the lower limits of flight altitude set by the automatic mission planning app to achieve finer ground sampling distances. The original images collected from the corn field had a resolution of 5472 × 3648 and a ground sampling distance (GSD) of 0.27 cm/pixel; while the original images collected from the soybean field had a resolution of 8192 × 5460 and a GSD of 0.06 cm/pixel. The highest GSD was achieved in 2022 with the 5 m altitude at which the impact on the canopy structure caused by the downwash airflow was minimized. Duplicate scenes, blurred images, and images with UAV shadow were excluded.

Table 1
Detailed information and parameters of data collection, raw images, and image pre-processing.

Crop	Corn		Soybean	
Date	July 2021		July 2022	
UAV and sensing system	Phantom 4 RTK with built-in RGB camera		Matrice 300 with Zenmuse P1 RGB camera	
Altitude AGL (m)	10		5	
GSD (cm/pixel)	0.27		0.06	
Resolution of original images	5472 × 3648		8192 × 5460	
Dataset	Dataset-1	Dataset-2	Dataset-1	Dataset-2
Number of original images used	15	10	31	10
Split ratio	8 × 8	10 × 7	4 × 3	16 × 10
Resolution of split sub-images	684 × 456	547 × 521	2048 × 1820	512 × 546
Resolution of resized sub-images	256 × 256	256 × 256	256 × 256	256 × 256

In this study, two datasets were prepared from the collected data: dataset-1 and dataset-2 (Table 1). Dataset-1 consisted of a total of 15 images (5472 × 3648 pixels) and 31 images (8192 × 5460 pixels) from UAV data collections of mid-growth-stage corn and soybeans infested with Palmer Amaranth. Dataset-1 was a dataset with existing pre-processing and labels that we did in a previous study. With a higher altitude used during the data collection (10 m AGL), raw images collected from the corn field in 2021 were split into 8 × 8 sub-images, each with a resolution of 684 × 456; while the raw images collected in 2022 were split into 4 × 3 sub-images, each with a resolution of 2048 × 1820 due to a lower altitude (5 m AGL) used during the data collection (Fig. 1). These split ratios were carefully selected to ensure enough spatial information in a sub-image for weed and crop classification. These sub-images were further resized as input images to the classification deep-learning models. Dataset-2 was pre-processed and labeled in the second phase, which consisted of 10 images (5472 × 3648 pixels) from the same corn fields and 10 images (8192 × 5460 pixels) from the same soybean fields. Slightly different split ratios were used for dataset-2. The original corn images were split into 10 × 7 sub-images, each with a resolution of 547 × 521; and the original soybean images were split into 16 × 10 sub-images, each with a resolution of 512 × 546.

Different resize ratios or resolutions of input images were tested. We found that, at a resolution of 256 × 256 for input images, models achieved the optimal combination of classification accuracy and inference speed. After removing some of the sub-images without any weeds in the scene, we resulted with 480 resized sub-images from corn fields and 360 resized sub-images from soybean fields for OD models (i.e., SSD and YOLOv8n); 120 resized sub-images from both crop fields for the DeepLabv3 + SS model; and 443 sub-images from corn fields and 353 sub-images from soybean fields for the proposed segmentation model MobileNetV4-Seg (Table 2). The resized images were labeled pixel-wise for SS modeling and with bounding boxes for OD modeling, respectively, using Computer Vision Annotation Tool (CVAT), which is an open-source data annotation platform. The training, validation, and test data split for all models are shown in Table 2.

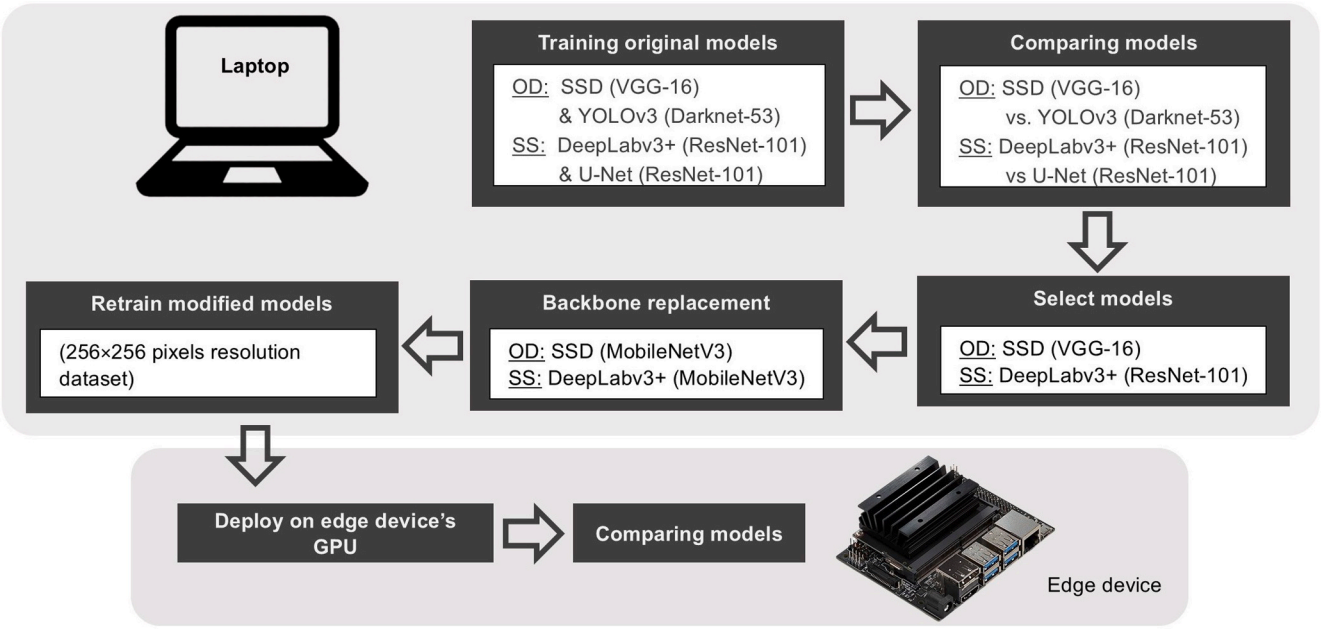


Fig. 1. Steps followed in implementing the weed detection models. OD in the figure stands for object detection, and SS stands for semantic segmentation.

Table 2
Total number of images and training, validation, and testing splitting for the object detection (OD) and semantic segmentation (SS) models.

Crop	Model	Total number of resized sub-images used	Number of training sub-images	Number of validation sub-images	Number of testing sub-images
Corn	Object Detection (OD)				
	SSD and YOLOv8n	480	400	40	40
	Semantic Segmentation (SS)				
	DeepLabv3+	120	100	10	10
	MobileNetV4-Seg	443	390	10	43
Soybean	Object Detection (OD)				
	SSD and YOLOv8n	360	300	30	30
	Semantic Segmentation (SS)				
	DeepLabv3+	120	100	10	10
	MobileNetV4-Seg	353	310	10	33

2.2. Overview of model selection and deployment process

For OD models, we considered two widely used ones, Single-Shot Detector (SSD) (W. Liu et al., 2016) and You Only Look Once (YOLOv3) (Redmon & Farhadi, 2018). We compared these two OD models on a laptop computer in terms of the IoU, F_1 score, and FPS, and found that the SSD model was better suited for our application (Table 4). For SS models we considered DeepLabv3+ (Yurtkulu et al., 2019)) and U-Net (Ronneberger et al., 2015), as they are two of the most commonly used models for semantic segmentation. We compared these two SS models on a laptop computer in terms of IoU, F_1 score, and FPS, and found that the DeepLabv3+ model performed better (Table 4). Both the selected OD (SSD) and SS (DeepLabv3+) models' original backbones were replaced by a lightweight backbone, MobileNetV3. These modified models were retrained with lower resolution (256 × 256) images and deployed on the edge device's GPU. Finally, the modified OD and SS models deployed on the edge device were compared between themselves in terms of detection or segmentation performance and inference speed. The methodology to work with the CNN based weed detection models is explained in a step-by-step manner in Fig. 1. We also evaluated additional two relatively newer lightweight models, the off-the-shelf YOLOv8n (Ultralytics, 2023) for OD and a proposed MobileNetV4-Seg for SS based on MobileNetV4 (Qin et al., 2024).

2.3. Model selection on laptop and desktop computers

The original SSD model came with a VGG-16 backbone (Tao et al., 2021) and the original DeepLabv3+ model came with a ResNet backbone (He et al., 2016). Since these original backbones contain a large number of parameters, they would not provide satisfied real-time inference if deployed directly on the Jetson Nano. Hence, the SSD's VGG-16 and the DeepLabv3+'s ResNet backbones were both replaced by the MobileNetV3 backbone (Howard et al., 2019) to efficiently run on a Jetson Nano for real-time inference, since the MobileNetV3 has a relatively small number of parameters. The architecture of the MobileNetV3 is shown in Fig. 2 (a). After replacing the original backbones, the modified SSD architecture with the MobileNetV3 backbone is shown in Fig. 2 (b) and the modified DeepLabv3+ architecture with the MobileNetV3 backbone is shown in Fig. 2 (c). Similarly, the proposed segmentation model, MobileNetV4-Seg comprises an encoder-decoder architecture for efficient semantic segmentation. The encoder module utilizes the feature extractor functionality of the pretrained MobileNetV4-Conv-Small (Qin et al., 2024) in the backbone to extract multi-scale features while maintaining computational efficiency. To enhance feature fusion and maintain spatial consistency, the extracted feature maps are up-sampled to the same resolution as the 1/8 feature map. A residual connection (skip connection) is then introduced to add the 1/8 feature map to the 1/32 feature map, avoiding gradient degradation and improving information propagation. In the decoder module, a dedicated semantic segmentation head (Lu et al., 2024) is employed to decode the extracted features into segmentation information. Finally, another semantic segmentation head processes the refined features, and the resulting segmentation map is up-sampled to match the original image size. MobileNetV4-Seg's architecture is depicted in Fig. 2(d).

Before being deployed on the edge devices, the initial model trainings of the modified lightweight SSD, YOLOv3, U-Net, and DeepLabv3+

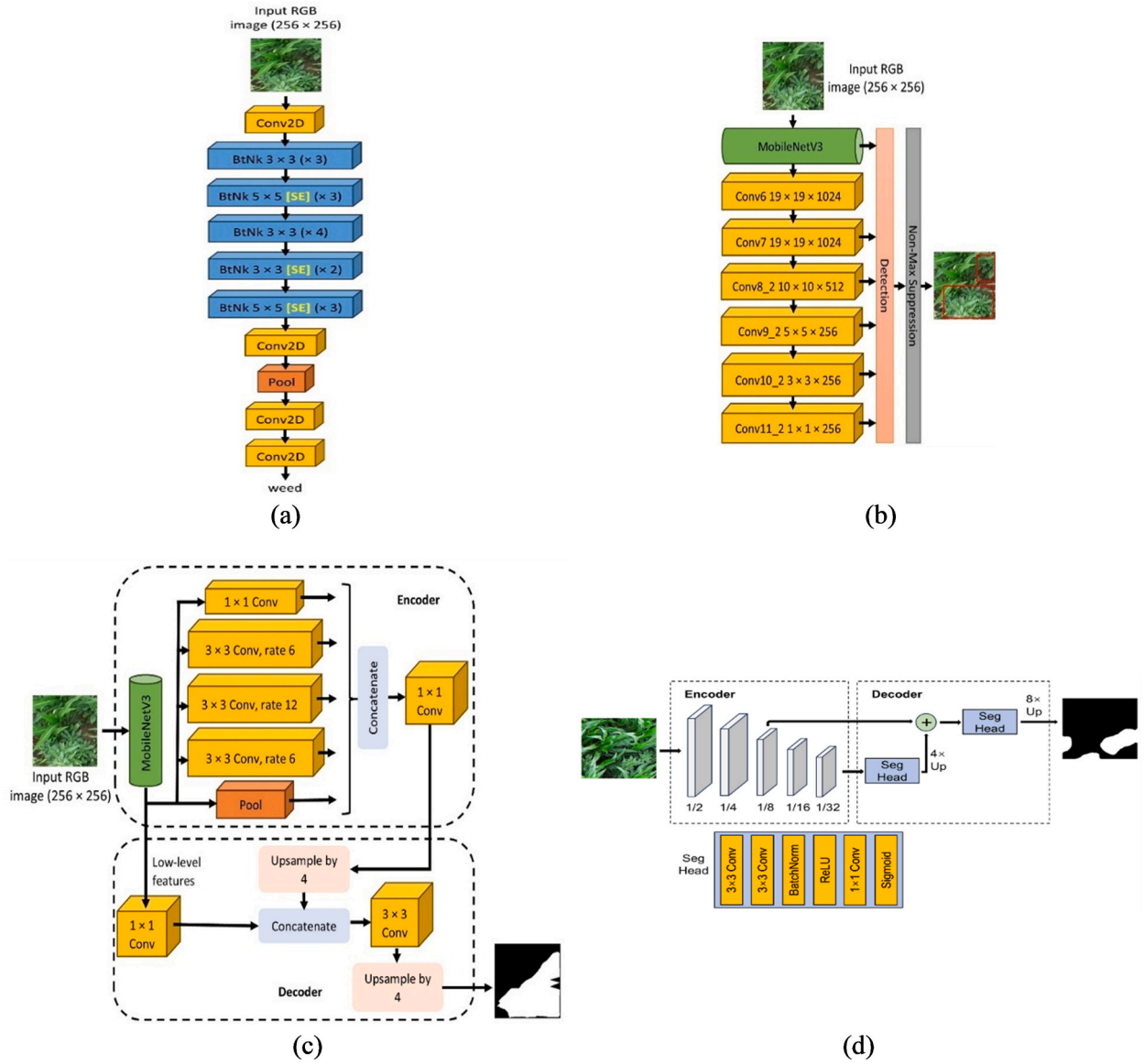


Fig. 2. Model architectures: (a) MobileNetV3 architecture adapted from Howard et al. (2019) and Nguyen et al. (2022). Conv2D is a convolutional layer, BtNk stands for bottleneck. SE denotes whether there is a Squeeze-And-Excite in that block. The × in the parentheses indicates how many times the block has been repeated. Pool represents the maxpooling layer. (b) Modified SSD architecture with MobileNetV3 backbone. Concept taken from Meng et al. (2022). (c) Modified DeepLabv3+ architecture with MobileNetV3 backbone adapted from Parajuli et al. (2023). (d) MobileNetV4-Seg architecture. The encoder module uses the down-sampling structure of MobileNetv4-Conv-Small (Qin et al., 2024) which down-samples the input image through five stages, generating feature maps at1/2, 1/4, 1/8, 1/16, and 1/32 dimensions. The decoder part uses a segmentation head (Lu et al., 2024) as shown at the bottom.

models were trained and evaluated on a laptop computer to utilize its better computational resource to speed up the process. The configurations of the laptop computer were CPU: 11th Gen Intel Core Processor i9-11950H @ 2.60 GHz; SSD: 1 TB, Core: 8, Cache: 24 MB, RAM: 32 GB; GPU: NVIDIA RTX A2000 Laptop GPU. PyTorch was used (torch version: 2.0.1, torchvision version: 0.15.2, and CUDA version: 12.5). Models with the minimum validation loss were selected. Similarly, in the second phase of the work, the YOLOv8n and MobileNetV4-Seg models were trained on a desktop computer (Processor: Intel$^{(R)}$ Core$^{(TM)}$ i7-10700F CPU @ 2.90 GHz, RAM: 32 GB, GPU: NVIDIA GeForce RTX 4090 24 GB GPU memory). The hyperparameters used during all the model training are provided in Table 3. The optimal parameters were selected based on the highest F_1 score of the validation dataset.

Table 3
Hyperparameters used during model training.

Model	Learning rate	Optimizer	Batch size	Number of epochs
SSD(original and modified)	0.001	SGD	32	300
YOLOv3	0.001	Adam	2	300
DeepLabv3+(original and modified)	0.0001	Adam	4	300
U-Net	0.0001	Adam	4	300
YOLOv8n	0.0001	Adam	20	300
MobileNetV4-Seg	0.0001	Adam	20	300

2.4. Model deployment on edge devices

NVIDIA Jetson Nano (GPU: NVIDIA Maxwell architecture with 128 NVIDIA CUDA cores, CPU: Quad-core ARM Cortex-A57 MPCore processor, RAM: 4 GB 64-bit LPDDR4, 1600 MHz 25.6 GB/s, storage: 16 GB eMMC 5.1, computing performance: 472 GFLOPS, weight: 250 g)

Table 4
Comparison of detection and segmentation performance, as well as inference speed, between OD models (SSD and YOLOv3), and between SS models (DeepLabv3+ and U-Net) for both corn and soybean datasets on PC.

Crop	Model	IoU (%)	F_1 score (%)	Inference speed (FPS)
		(OD)		
Corn	SSD	64.3	75.8	72
	YOLOv3	47.7	55.2	90
Soybean	SSD	48.3	68.9	72
	YOLOv3	46.4	40.0	90
		(SS)		
Corn	DeepLabv3+	61.0	77.3	50
	U-Net	53.0	72	40
Soybean	DeepLabv3+	65.5	82.7	50
	U-Net	62.0	81.4	40

(NVIDIA, 2024) and Jetson Orin Nano (1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores; 6-core Arm Cort4ex-A78AE v8.2 64-bit CPU; 8 GB 128-bit LPDDR5) were the two resource-constrained edge devices explored in this study. CNN models were deployed on Jetson Nano's GPU utilizing PyTorch with torch version: 2.4.1, torchvision version: 0.19.1, and CUDA version: 11.4; and Jetson Orin Nano's GPU utilizing PyTorch but with torch version: 1.10.0, torchvision version: 0.11.3, and CUDA version: 10.2. SSD and DeepLabv3+ models were deployed on Jetson Nano, and YOLOv8n and MobileNetV4-Seg models were deployed on Jetson Orin Nano. Due to differences in the operating system architectures between the PC and the Jetson platform, the versions of software packages supported on the PC may not always be compatible with the Jetson system. In order to deploy the model originally developed with the laptop or desktop onto the Jetson devices, we often needed to install additional necessary supporting packages for the Jetson system.

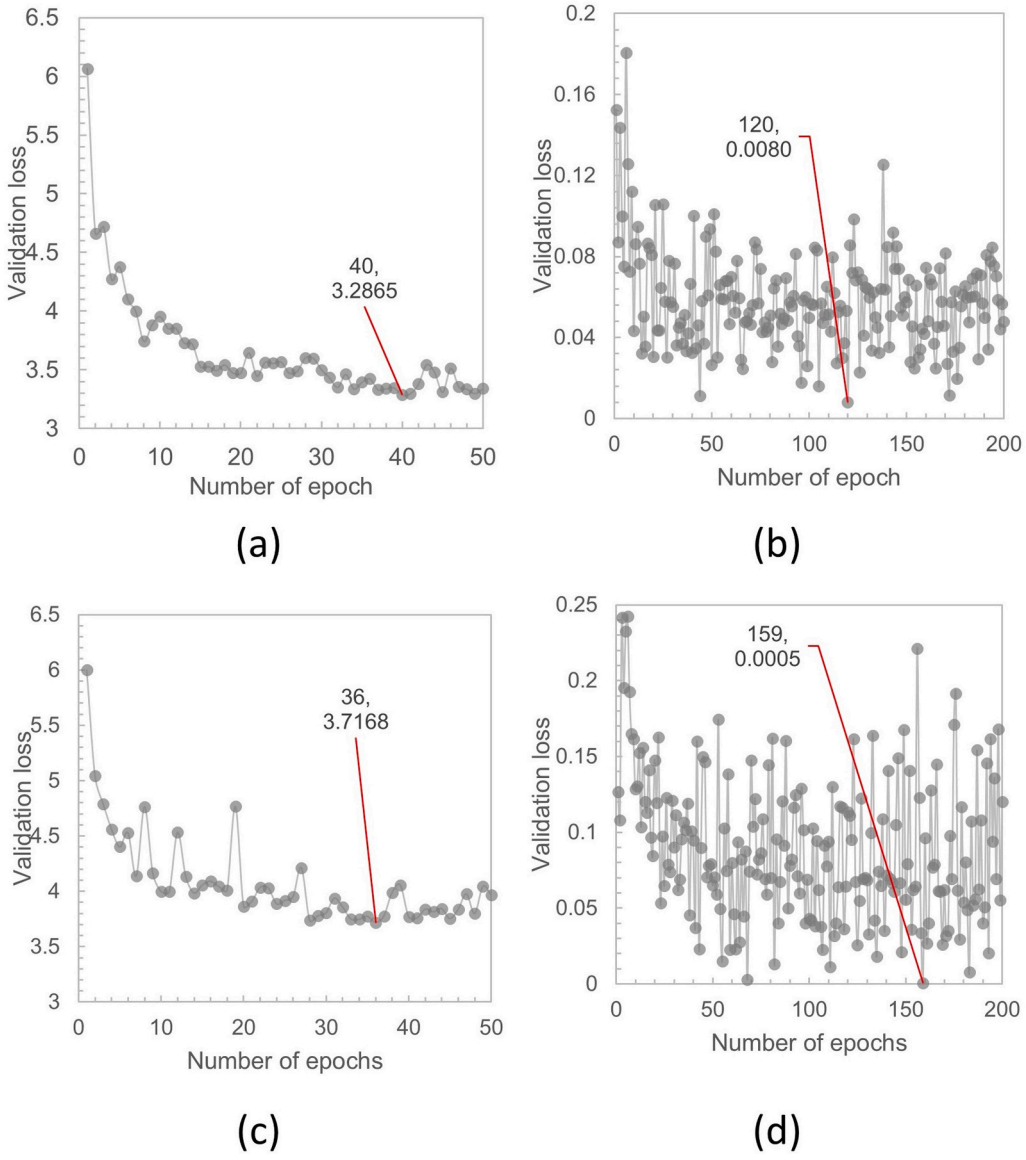


Fig. 3. Number of epochs versus validation loss for: (a) the SSD model for the corn dataset, (b) DeepLabv3+ model for the corn dataset, (c) the SSD model for the soybean dataset, (d) DeepLabv3+ model for the soybean dataset. The point with the lowest validation loss is marked in red. For (a, c) only 50 out of 300 epochs, and for (b, d) only 200 out of 300 epochs are shown for better visualization. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.5. System performance evaluation

In this study, we used metrics including IoU (with a threshold of 0.5), precision, recall, and F_1 score (Eqs. (1)-(4)) to evaluate the model performance on weed detection or segmentation. IoU is the ratio between the overlap (or intersection) and the union, of the ground truth bounding box (BB_{GT}) and the prediction bounding box (BB_{pred}). For OD, if the IoU for a particular object was equal to or greater than 0.5, then it was considered as a true positive, else false positive. Other parameters such as True Positive (*TP*), True Negative (*TN*), False Positive (*FP*), False Negative (*FN*) are explained below.

$$IoU = \frac{Area(BB_{pred} \cap BB_{GT})}{Area(BB_{pred} \cup BB_{GT})} \quad (1)$$

$$Precision(Pr) = \frac{TP}{TP + FP} \quad (2)$$

$$Recall(Rc) = \frac{TP}{TP + FN} \quad (3)$$

$$F_1\ score = \frac{2 \times Pr \times Rc}{Pr + Rc} \quad (4)$$

3. Results

3.1. Selected OD and SS models with original backbones on PC

Fig. 3 shows the number of epochs versus validation loss plots for SSD and DeepLabv3+ models deployed on the laptop computer. Models with the lowest validation loss were selected for the next step (i.e. performance comparison). YOLOv3 and U-Net's plots are not shown here as they were outperformed by the other two models (refer to Table 4). Table 4 shows the detection performance (IoU and F_1 score) and inference speed comparison of the two OD models on the laptop computer. From Table 4 we can see that, although YOLOv3 has a better inference speed compared to SSD, SSD has a better IoU and F_1 score with a reasonable inference speed. For that reason, SSD was selected as the better OD model. The segmentation performance and inference speed comparison of the two SS models on the laptop computer are also shown in Table 4. As shown in Table 4, DeepLabv3+ achieved better IoU, F1 score, and inference speed compared to U-Net in this case.

Between SSD and YOLOv3, we selected SSD for further deployment on the edge device, since SSD achieved a higher IoU and F_1 score with reasonable inference speed on our datasets, despite YOLOv3 offering faster inference speed. Similarly, DeepLabv3+ was selected over the U-Net in terms of IoU, F_1 score, and inference speed. DeepLabv3+ uses an advanced technique called atrous (dilated) convolutions, which helps it capture information at multiple scales. This improves the segmentation performance of DeepLabv3+ on objects of different sizes, and hence it was selected as the preferred weed detection SS model. Then we modified the SSD and DeepLabv3+ models, deployed them on the Jetson Nano, and tested their performance for weed in the corn and soybean dataset.

3.2. Performance comparison of lightweight OD and SS models on edge devices

The selected OD and SS models with better performance on the laptop computer from the previous subsection, i.e., the SSD and the DeepLabv3+ models, were modified and deployed on the Jetson Nano and tested with unseen resized sub-images in 256 × 256 pixel resolution. Similarly, the YOLOv8n (OD model) and the proposed MobileNetV4-Seg (SS model) were deployed on the Jetson Orin Nano. The performance metrics (IoU, Precision, Recall, and F_1 score) and inference speeds are shown in Table 5.

The lightweight SSD model achieved a real-time inference speed of 33 FPS for 256 × 256 pixel images when deployed on Jetson Nano. Performance metrics for the SSD model included an IoU of 58.8 %, precision of 62.5 %, recall of 95.2 %, and an F_1 score of 75.5 % for corn. For soybean, it attained 42.3 % IoU, 63.6 % precision, 77.8 % recall, and a 70 % F_1 score. As for the lightweight DeepLabv3+ model operated at a near real-time inference speed of 10 FPS for 256 × 256 pixel images on Jetson Nano. It achieved an IoU of 60.0 %, precision of 82.0 %, recall of 72.6 %, and an F_1 score of 77.0 % for corn. For soybean, it recorded 62.4 % IoU, 81.5 % precision, 82.0 % recall, and an 81.7 % F_1 score. With Jetson Orin Nano, YOLOv8n and the MobileNetV4-Seg both achieved real-time inference speeds of 30 FPS and 44 FPS, respectively, for 256 × 256 pixel images. The YOLOv8n model recorded an IoU of 58.6 %, precision of 87.0 %, recall of 81.6 %, and an F_1 score of 84.2 % for corn, while for soybean, it achieved 56.1 % IoU, 77.1 % precision, 72.7 % recall, and an F_1 score of 74.9 %. The MobileNetV4-Seg model achieved an IoU of 69.9 %, precision of 80.9 %, recall of 83.7 %, and an F_1 score of 82.3 % for corn. For soybean, it achieved 76.8 % IoU, 83.8 % precision, 90.2 % recall, and an F_1 score of 86.9 %.

Table 5
Detection or segmentation performances and inference speeds of the OD and the SS models on edge devices (SSD and DeepLabv3+ on Jetson Nano, YOLOv8n and MobileNetV4-Seg on Jetson Orin Nano).

Model	IoU (%)	Precision (%)	Recall (%)	F_1 score (%)	Inference speed (FPS)
		(Corn)			
Lightweight SSD (OD)	58.8	62.5	95.2	75.5	33
YOLOv8n (OD)	58.6	87.0	81.6	84.2	30
Lightweight DeepLabv3+ (SS)	60.0	82.0	72.6	77.0	10
MobileNetV4-Seg (SS)	69.9	80.9	83.7	82.3	44
		(Soybean)			
Lightweight SSD (OD)	42.3	63.6	77.8	70.0	33
YOLOv8n (OD)	56.1	77.1	72.7	74.9	30
Lightweight DeepLabv3+ (SS)	62.4	81.5	82.0	81.7	10
MobileNetV4-Seg (SS)	76.8	83.8	90.2	86.9	44

3.2.1. Performance comparison on data collected in corn

Images were randomly selected to show the performance of each model on the data collected in corn and soybean (Figs. 4–9). Performance of the three OD models in the corn datasets is shown in Fig. 4. The lightweight SSD had lower Precision but higher Recall than the YOLOv8n (Table 5). In Fig. 4(d), a false positive can be observed. Some of the weeds were missed in being labeled as it was not clearly visible to human eyes, but the model detected them. On the right side of the image, instead of detecting the whole patch of weed, the model detected two separate weeds. The reason is there are human errors and differences in how the images are labeled. Spread out weeds can be labeled using bounding boxes in several ways, and thus detection may not exactly match with the ground truth. In Fig. 4(e), there is a false positive similar to Fig. 4(d). On the bottom right of the image, the model detected weeds well. However, the model did not detect the weed above it, as it was covered by the corn. Fig. 4(f), (g), and (h) gave almost similar results to SSD. For Fig. 4(i) and (j), YOLOv8n detected only the sharp weed images and missed the subtle presence of weeds, which SSD detected. This way YOLOv8n had better results than SSD as these subtle weed images were not labeled in ground truth and hence considered as false positives for SSD.

Figs. 5 and 6 show the segmentations using DeepLabv3+ and MobileNetV4-Seg SS models for five randomly selected images collected from the corn fields. Fig. 5(k) does not exactly match with the ground

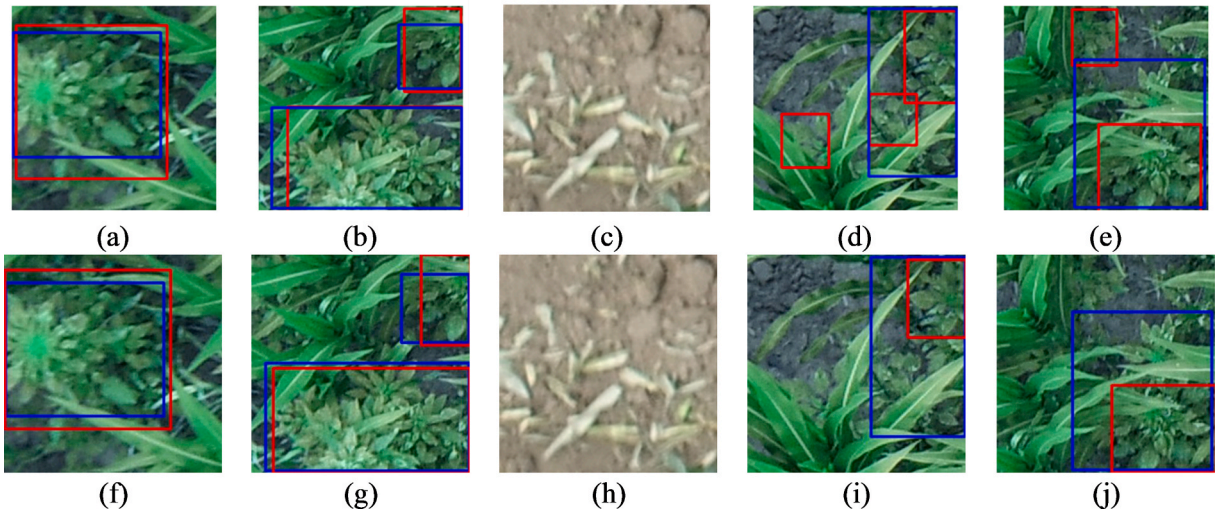


Fig. 4. Five example cases of Palmer Amaranth weed detection in corn using the lightweight SSD (a-e) and YOLOv8n (f-j) OD model. Red boxes represent the detection of weeds by the model, and blue boxes represent the ground truth. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

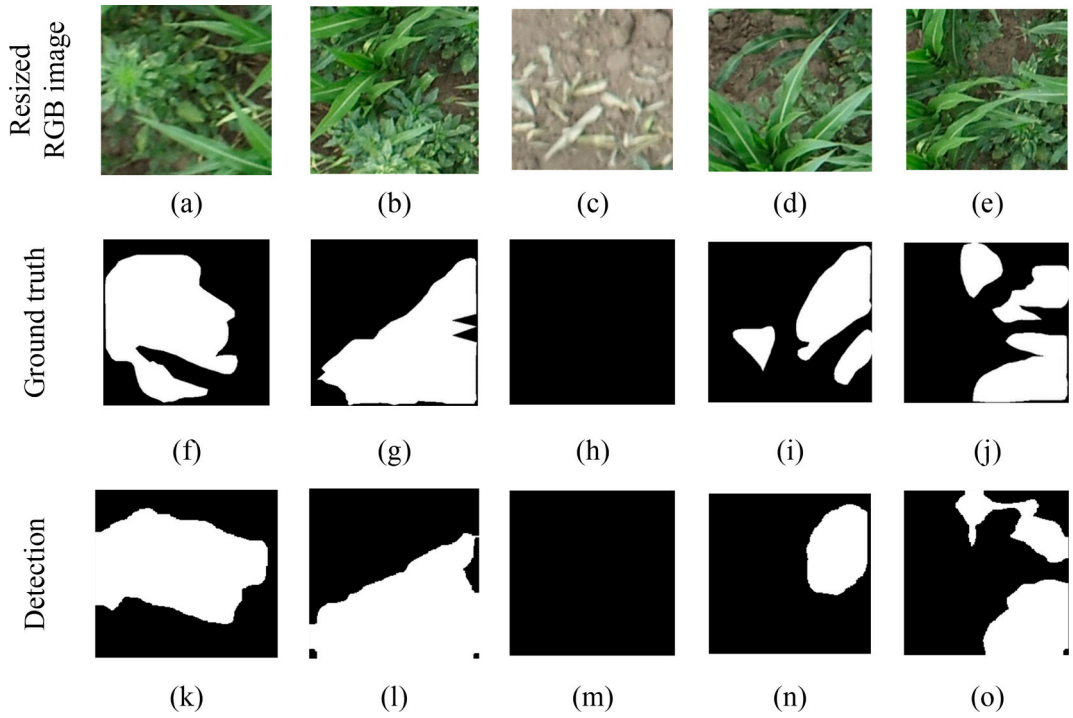


Fig. 5. Five example cases of Palmer Amaranth weed detection in corn using the lightweight DeepLabv3 + SS model. (a-e) are the resized RGB sub-images, (f-j) are the ground truth masks, and (k-o) show the detections using the SS model. For the ground truth mask and detection, white indicates the weeds, while black represents the background.

truth, Fig. 5(f). This is due to the fact that at one place there are weeds underneath the corn. During the labeling, the corn part is excluded from the weeds, but the model detected the whole weed segment. For Fig. 5 (b) and Fig. 5(c) the model performs really well for detecting if there is any weed present or not. Fig. 5(d) has the similar issue of corn covering the weed as Fig. 5(a), hence the detection, Fig. 5(n) and the ground truth, Fig. 5(i) do not exactly match. Except for some false detections, the detection, Fig. 5(o) is close enough to the ground truth, Fig. 5(j) for Fig. 5(e). In Fig. 6(a), MobileNetV4-Seg detected weeds in the same region as the ground truth. Whereas DeepLabv3+ also detected an additional surrounding area of the ground truth as weed. For Fig. 6(b), MobileNetV4-Seg detected the soil in the middle right of the image as background, which was detected as weed by DeepLabv3+. In Fig. 6(c), no weed was present, hence not detected. For Fig. 6(d) and (e), MobileNetV4-Seg's detections are closer to the ground truth compared to DeepLabv3+'s detections.

3.2.2. *Performance comparison on data collected in soybean*

Similarly, images were randomly selected for the data collected from the soybean fields. Fig. 7 shows the results of five images with the SSD and YOLOv8n OD models. In Fig. 7(a) and Fig. 7(b), the whole image scenes were covered by weeds. Although the whole patch of weeds was detected, both images had additional sub-detections. In Fig. 7(c), the model performs well as it detected a few weeds that were missed in the

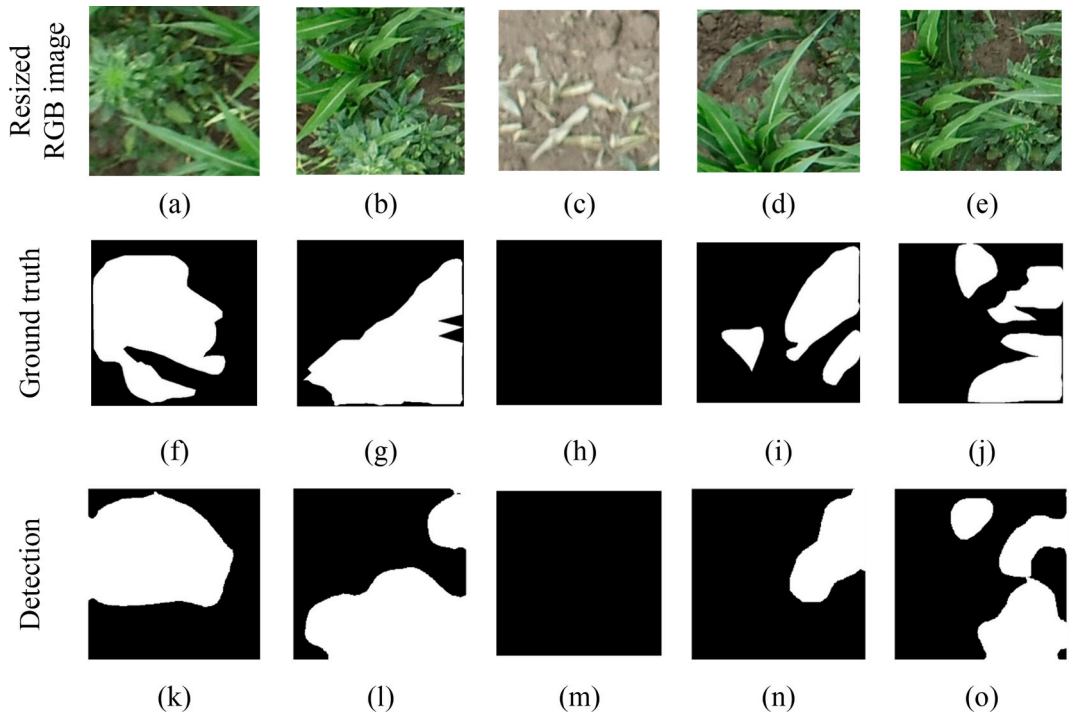


Fig. 6. Five example cases of Palmer Amaranth weed detection in corn using the MobileNetV4-Seg SS model. (a-e) are the resized RGB sub-images, (f-j) are the ground truth masks, and (k-o) show the detections using the SS model. For the ground truth mask and detection, white indicates the weeds, while black represents the background.

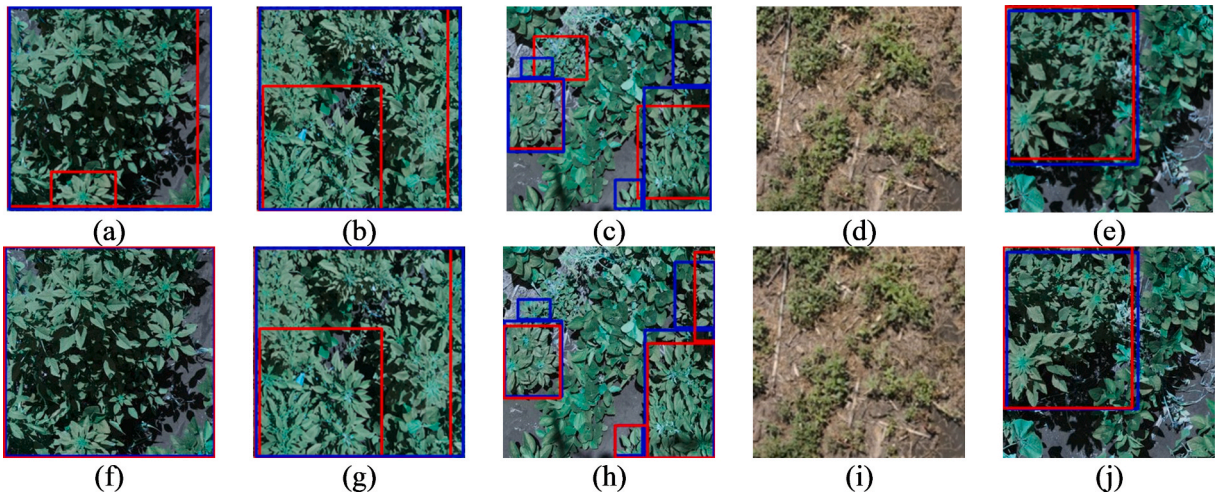


Fig. 7. Five example cases of Palmer Amaranth weed detection in soybean using the lightweight SSD (a-e) and YOLOv8n (f-j) OD models. Red boxes represent the detection of weeds by the model, and blue boxes represent the ground truth. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

original labeling; however, the model missed the detection at the top right corner. In Fig. 7(d), the weeds shown in the scene were not Palmer Amaranth and hence were not detected by either model. The models performed well for the cases in Fig. 7(e) and 7(j). For scenes shown in Fig. 7(g), (i), and (j), YOLOv8n's detections agreed with SSD. For Fig. 7 (h), YOLOv8n captured almost all weed patches including the top right and bottom weed patches; however, it did not detect the upper left patch that was missing in the labeling compared to SSD.

Examples of DeepLabv3+ and MobileNetV4-Seg SS models' detections are shown in Figs. 8 and 9, respectively. For Fig. 8(a), the detection, Fig. 8(k) and the ground truth, Fig. 8(f) are almost identical. For Fig. 8(b), the whole image was labeled as covered by weeds, but actually there are some bare soil in some small places. These very small weed free locations were also detected by the model, so in that case, the model performed better than the human. In Fig. 8(c), at the center of the image, the model detects a few soybeans as weeds (false positives). For Fig. 8(d), there is no weed labeled in Fig. 8(i) and the model does a great job for the prediction in Fig. 8(n). Fig. 8(e) also has some false positives in the detection as shown in Fig. 8(o) similar to Fig. 8(c), because it is difficult to tell if these false positives are actually weed or soybean from the low resolution images even though they are not labeled as weeds in the ground truth in Fig. 8(j). For Fig. 9(a), (c), and (e), the detections are almost identical to the ground truths. For Fig. 9(b) and (d), it can be said that MobileNetV4-Seg performed better than the ground truth,

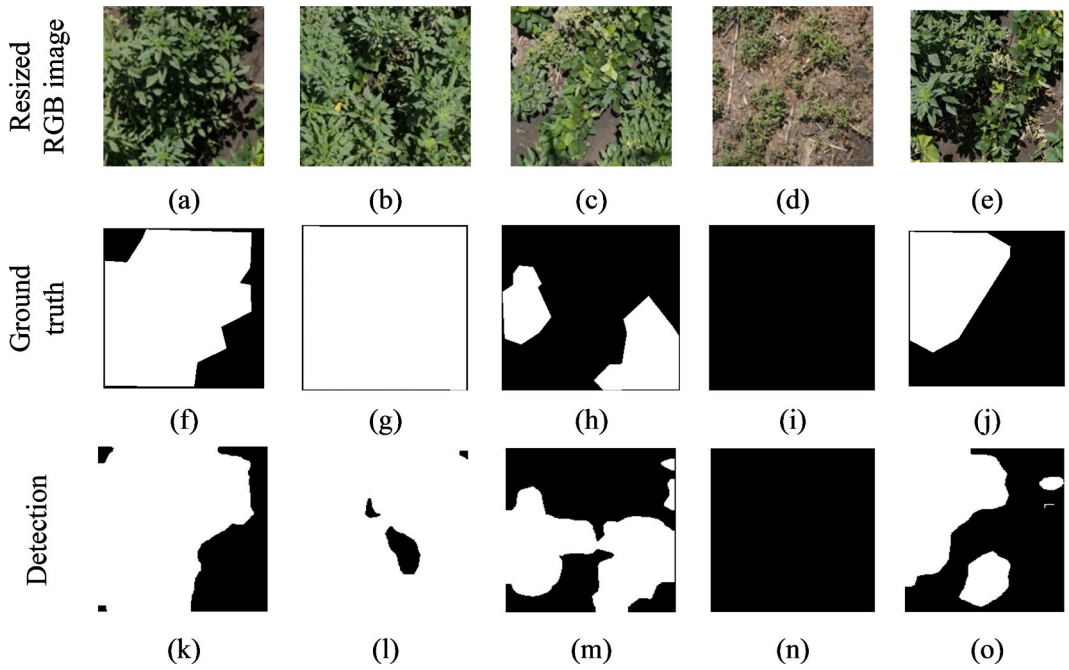


Fig. 8. Five example cases of Palmer Amaranth weed detection in soybean using the lightweight DeepLabv3+ semantic segmentation model. (a-e) show the actual images, (f-j) show the ground truth masks, and (k-o) show the detections using the SS model. For the ground truth mask and detection, white indicates the weeds, while black represents the background.

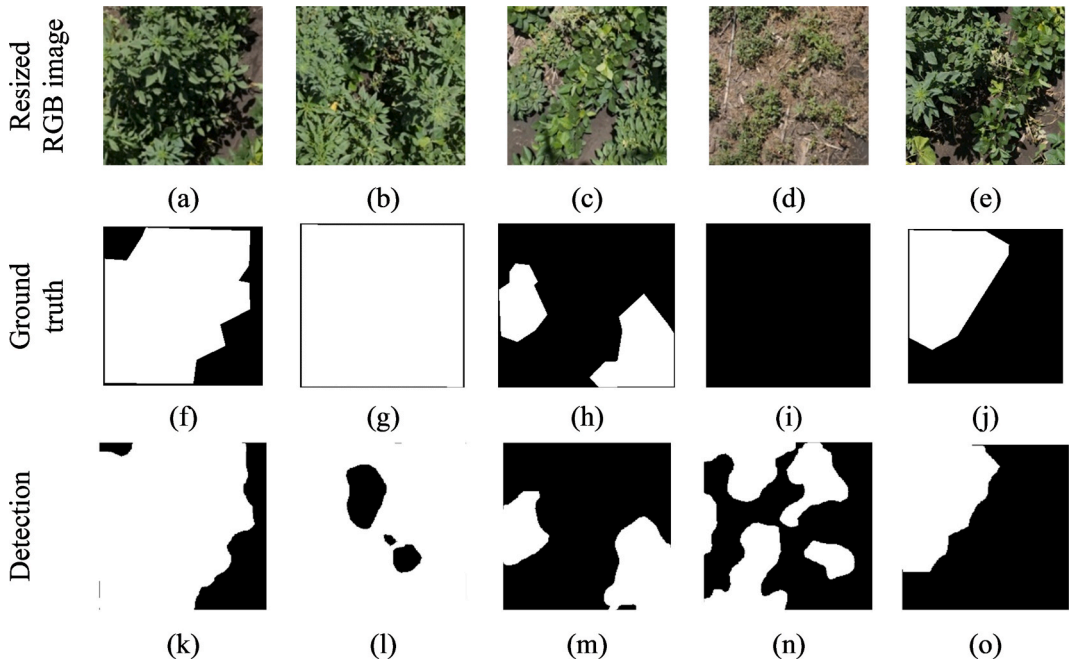


Fig. 9. Five example cases of Palmer Amaranth weed detection in soybean using the MobileNetV4-Seg semantic segmentation model. (a-e) show the actual images, (f-j) show the ground truth masks, and (k-o) show the detections using the SS model. For the ground truth mask and detection, white indicates the weeds, while black represents the background.

considering the human error in the labeling. In Fig. 9(b), there is some background in the middle of the image, which was not labeled in the ground truth, but was detected by the model. Similarly, Fig. 9(d) was labeled as all background. However, the small weeds were detected by the model. By observing the Figs. 4-9, it can be suggested that using large dataset (for training, validation, and testing) and being consistent in labeling throughout the dataset are helpful to improve the performance of the models.

4. Discussion

While extensive work has been carried out using machine or deep learning models for automatic weed detection, many of these efforts have focused on improving detection accuracy through the development of advanced model structures. However, adapting these models for deployment on edge devices, which have limited computational power but require ultra-fast inference speeds, particularly for real-time aerial

applications such as those based on spray UAVs or spray drones, remains a significant challenge. In this research, we demonstrated the process and performance of developing and deploying lightweight computer vision models on two of the most widely used resource-constrained edge computing devices.

4.1. Model deployment on edge devices

We deployed modified lightweight weed detection models on two resource-constrained edge devices, both of which are equipped with GPUs for efficient parallel processing of computational tasks. Their CPUs support multi-threaded processing, which is essential for handling the multiple tasks required during model deployment. Use the Jetson Nano we used as an example, it has a computing performance of 472 GFLOPS and a lightweight design at just 250 g, which makes it well-suited for edge deployments where both performance and portability are critical, such as UAV-based applications. Similarly, the Jetson Orin Nano is the latest model and computationally more resourceful with 1280 GFLOPS for FP32 precision. Ensuring the models are lightweight so that the RAM on the edge device can provide sufficient memory resources for running the model is crucial.

We used PyTorch for model deployment on edge devices, utilizing specific versions of software packages to ensure compatibility with Jetson's architecture. It is important to note that the architectural differences between the Jetson devices and a laptop or desktop PC can lead to compatibility issues. Specifically, certain software packages that are supported and function properly on a laptop or desktop PC may not be directly supported by the Jetson devices due to their unique architecture. In many instances, additional support packages or modifications to the existing code are required to ensure the reliable and efficient deployment of the models, allowing them to run smoothly on Jetson devices.

All inference speeds mentioned in this work refer solely to the time required to process an image through the model. The time required to load the image from the device's storage was not included, though in our case it was trivial. Additionally, there is a "model loading time" associated with the inference, which refers to the time required to load the model onto the GPU. This causes the inference of the first couple of images to take relatively longer times, but reduces once the model is loaded. Another factor is the time required to acquire images from the camera in the on-the-go application. All these factors could vary and impact the overall system speed, and need to be considered in applications.

4.2. Lightweight OD and SS models for real-time applications

In this study, some of the models we evaluated are lightweight as they are, i.e., the YOLOv8n object detection model and the MobileNetV4-Seg semantic segmentation model; some of the models, i.e., the SSD and YOLOv3 object detection models and the U-Net and DeepLabv3+ semantic segmentation models, were initially too heavy to be run on a resource-constrained edge device and needed to be simplified and optimized before deployment. For models in the latter case, we did not go the route of using a model optimizer such as TensorRT, but replaced the original heavy backbones of the model with a pre-trained lightweight MobileNetV3 backbone. Results showed acceptable accuracy with real-time performance.

When comparing the OD models used in this study by themselves, i.e., the modified lightweight SSD and the YOLOv8n, the YOLOv8n did not outperform the SSD model in every case, and it also had a lower FPS compared to the modified SSD. When comparing the SS models, i.e., the modified lightweight DeepLabv3+ and the proposed MobileNetV4-Seg, it can be observed that the proposed MobileNetV4-Seg exhibits superior performance to the DeepLabv3+ in most cases and also achieves a significantly improved FPS.

In this study, both object detection and semantic segmentation achieved comparable real-time performance (Table 5) with the particular input image size we used (256 × 256 pixels), which exceeded our initial expectation, as we typically consider object detection to be superior for real-time applications. The OD models achieved up to 33 FPS on the Jetson Nano, while the SS models achieved up to 44 FPS of inference time on a slightly more powerful Jetson Orin Nano, all with comparable accuracies. Although YOLOv8n (OD model) performed better than MobileNetV4-Seg (SS model) in some cases for the corn dataset, MobileNetV4-Seg consistently outperformed YOLOv8n for the soybean dataset. This may be partially due to the higher image resolution of the soybean datasets, but the efficient model structure of both models also plays a significant role, especially given the challenge of distinguishing between Palmar Amaranth and soybeans, which appear similar to each other.

The proposed MobileNetV4-Seg also showed faster inference speed compared to YOLOv8n, as our developed model has only 1.93 million parameters, whereas YOLOv8n has 3.2 million parameters (Ultralytics, 2023). The MobileNetV4-Conv-Small encoder of the MobileNetV4-Seg efficiently captures multi-scale features with low computational overhead, and the semantic segmentation head in the decoder combined with residual connections enhances spatial consistency and gradient flow, leading to better localization of object boundaries. Despite the different spatial variation and scale differences introduced by these new samples, the model maintained high accuracy for weed segmentation. This indicates the model's robustness and generalization capability to dataset variations, making it suitable for diverse real-world applications.

4.3. Limitations and future work

This study has several limitations that should be acknowledged in order to clarify the scope and applicability of the results. First, the size and complexity of the dataset used in model training and evaluation were limited. Specifically, the datasets used for OD models (SSD and YOLOv8n) were 480 and 360 sub-images for corn and soybean, respectively, while the dataset for DeepLabv3+ SS model was 120 sub-images for both corn and soybean, and datasets for MobileNetV4-Seg were 443 sub-images for corn and 353 sub-images for soybean. Although these datasets were sufficient for preliminary validation, such sample sizes may not capture the full range of variability in field conditions. Factors such as weed shape variation, lighting changes, occlusion, soil background diversity, and plant density can greatly affect model robustness. Expanding the dataset by including images from different locations, times, and incorporating augmented and synthetic data could significantly improve generalizability.

Second, the Jetson Nano was initially selected due to its low power consumption, GPU acceleration, and its potential for lightweight UAV integration. However, its low processing power and memory limited the model complexity that could be deployed on it in real time. For instance, DeepLabv3+ achieved only 10 FPS for 256 × 256 image inference, which may be insufficient for fast-moving UAV-based applications. Although the resourceful device, Jetson Orin Nano allowed deployment of newer models like YOLOv8n and MobileNetV4-Seg, the study did not explore deployment on other edge platforms or mini PCs having powerful CPUs without GPUs, such as the Intel NUC or Google's Coral Dev Board, which may be capable of supporting quantized or pruned versions of tiny deep learning models. Evaluation on these alternatives could provide insights into cost-effective, GPU-free deployments suitable for constrained field settings.

Third, the study focuses on detecting Palmer Amaranth weeds only in corn and soybean fields during specific crop growth stages. It limits the system's applicability to broader scenarios as other common weed species (e.g., waterhemp, kochia, morning glory) and complex field conditions such as overlapping weed and crop canopies, mixed-species weed patches, or partial occlusion were not included. Furthermore, data collection was done under relatively stable lighting and soil background conditions. The lack of variation in crop stages (e.g., early

vegetative vs late reproductive) also limits the model's generalization. These gaps could affect detection accuracy in real-world deployments and warrant further exploration.

Finally, while this work contributes to a larger UAV-based precision spraying project, the current study is limited to model training, evaluation, and deployment on edge devices. It does not include the integration of the vision system with UAV flight controllers and its performance evaluation under real aerial operating conditions. Important environmental factors such as UAV vibration, motion blur, and wind effect can influence both image quality and real-time detection performance. Incorporating these elements in future studies will be crucial to validate the robustness of the system in dynamic field conditions. In future work, we aim to conduct full integration of the vision system with the flight system and perform the flight-based validation of the autonomous system.

5. Conclusion

In this study, we successfully developed and deployed four lightweight CNN OD and SS models on resource-constrained edge devices, i.e., Jetson Nano and Jetson Orin Nano, for real-time detection and segmentation of Palmar Amaranth in mid-growth stage corn and soybeans. With input image size of 256 × 256 pixels, the modified lightweight SSD OD model achieved real-time (33 FPS) inference speed on Jetson Nano, with 58.8 % IoU, 62.50 % precision, 95.2 % recall, and 75.5 % F_1 score for corn, and 42.3 % IoU, 63.60 % precision, 77.8 % recall, and 70.0 % F_1 score for soybean. The modified lightweight DeepLabv3+ SS model achieved near-real-time (10 FPS) inference speed on Jetson Nano, with 60.0 % IoU, 82.0 % precision, 72.6 % recall, and 77.0 % F_1 score for corn, and 62.4 % IoU, 81.5 % precision, 82.0 % recall, and 81.7 % F_1 score for soybean. The un-modified YOLOv8n OD model achieved real-time (30 FPS) inference speed on Jetson Orin Nano, with 58.6 % IoU, 86.96 % precision, 81.6 % recall, and 84.2 % F_1 score for corn; 56.1 % IoU, 77.1 % precision, 72.8 % recall, and 74.9 % F_1 score for soybean. The proposed MobileNetV4-Seg SS model achieved real-time (44 FPS) inference speed on Jetson Orin Nano, with 69.9 % IoU, 80.9 % precision, 83.7 % recall, and 82.3 % F_1 score for corn, and 76.8 % IoU, 83.8 % precision, 90.2 % recall, and 86.9 % F_1 score for soybean. All deployments resulted in acceptable detection and segmentation accuracies and inference speed, except for the modified DeepLabv3+ was a little slow for our case. The semantic segmentation models, such as the MobileNetV4-Seg model, had comparable if not superior performance compared to state-of-the-art object detection models. The models with top performance can be further improved and integrated into the on-the-go systems.

CRediT authorship contribution statement

Md Didarul Islam: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Wenxin Liu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Pascal Izere:** Writing – review & editing, Data curation. **Puranjit Singh:** Writing – review & editing, Software, Data curation. **Chong Yu:** Writing – review & editing, Methodology. **Benjamin Riggan:** Writing – review & editing, Methodology. **Kuan Zhang:** Writing – review & editing, Methodology. **Amit J. Jhala:** Writing – review & editing, Resources, Methodology. **Stevan Knezevic:** Writing – review & editing, Methodology. **Yufeng Ge:** Writing – review & editing, Methodology. **Santosh Pitla:** Writing – review & editing, Methodology. **Joe Luck:** Writing – review & editing, Methodology. **Yeyin Shi:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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